Date: 11th March 2026 DAB-II Section:A

Topic to be cover: Hypothesis testing of mean

Hypothesis testing of mean

There are 6 basic steps of hypothesis testing for population mean:

1) State the claim & opposite to the claim
2) Level of Significance
3) Test Statistics
4) Critical region
5) Calculation
6) Conclusion

Roadmap of mean testing

<table>
<tr><td>Hypothesis</td><td>Test Statistics</td><td>Critical region</td></tr>
<tr><td>H_1: μ>a</td><td rowspan="3">$z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$
When population variance/ population Standard deviation is known
$t = \frac{\bar{X}-\mu}{s/\sqrt{n}}$
When population variance/ population Standard deviation is unknown</td><td>$Z>Z_{\alpha}$
$t>t_{\alpha,n-1}$
If critical region satisfies reject Ho</td></tr>
<tr><td>H_1: μ<a</td><td>$Z<-Z_{\alpha}$
$T<-t_{\alpha,n-1}$
If critical region satisfies reject Ho</td></tr>
<tr><td>H_1: μ≠a</td><td>$Z>Z_{\alpha/2}$ or $Z<-Z_{\alpha/2}$
$t>t_{\alpha/2,n-1}$ or $t<-t_{\alpha/2,n-1}$
If critical region satisfies reject Ho</td></tr>
</table>

Exercise 8.3 , Book Elementary statistics by bluman

Question#8**Commute Time to Work** A survey of 15 large U.S. cities finds that the average commute time one way is 25.4 minutes. A chamber of commerce executive feels that the commute in his city is less and wants to publicize this. He randomly selects 25 commuters and finds the average is 22.1 minutes with a standard deviation of 5.3 minutes. At α=0.10, is he correct?

Step#1

Ho:

H_1:

Step#2

Level of significance: α

Step#3

Test Statistics

$$z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

$$t = \frac{\overline{X} - \mu}{s/\sqrt{n}}$$

Step#4

Critical region

$Z > Z_{\alpha}$

$t > t_{\alpha,n-1}$

$Z < -Z_{\alpha}$

$t < -t_{\alpha,n-1}$

$Z > Z_{\alpha/2}$ or $Z < -Z_{\alpha/2}$

$t > t_{\alpha/2,n-1}$ or $t < -t_{\alpha/2,n-1}$

If critical region satisfies reject Ho

Step#5

Calculation:

$\mu =, s =, \overline{X} =, n =,$

$$t = \frac{\overline{X} - \mu}{s/\sqrt{n}}$$

Step#6

Conclusion:

Question#9 **Heights of Tall Buildings** A researcher estimates that the average height of the buildings of 30 or more stories in a large city is at least 700 feet. A random sample of 10 buildings is selected, and the heights in feet are shown. At $\alpha = 0.025$, is there enough evidence to reject the claim?

485, 511, 841, 725, 615, 520, 535, 635, 616 & 582

Step#1

Ho:

H_1:

Step#2

Level of significance: α

Step#3

Test Statistics

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

Step#4

Critical region

$Z > Z_\alpha$

$t > t_{\alpha,n-1}$

$Z < -Z_\alpha$

$t < -t_{\alpha,n-1}$

$Z > Z_{\alpha/2}$ or $Z < -Z_{\alpha/2}$

$t > t_{\alpha/2,n-1}$ or $t < -t_{\alpha/2,n-1}$

If critical region satisfies reject Ho

Step#5

Calculation:

μ=, $s\ =,\ \overline{X}\ =,\ n\ =,$

$$t\ = \frac{\overline{X}-\mu}{s/\sqrt{n}}$$

Step#6

Conclusion:

Question# 12. **Internet Visits** AU.S. Web Usage Snapshot indicated a monthly average of 36 Internet visits per user from home. A random sample of 24 Internet users yielded a sample mean of 42.1 visits with a standard deviation of 5.3. At the 0.01 level of significance can it be concluded that this differs from the national average?

Step#1

Ho:

H_1:

Step#2

Level of significance: α

Step#3

Test Statistics

$$z\ = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$$

$$t\ = \frac{\overline{X}-\mu}{s/\sqrt{n}}$$

Step#4

Critical region

$Z>Z_{\alpha}$

$t>t_{\alpha,n-1}$

$Z<-Z_{\alpha}$

$t<-t_{\alpha,n-1}$

$Z>Z_{\alpha/2}$ or $Z<-Z_{\alpha/2}$

$t>t_{\alpha/2,n-1}$ or $t<-t_{\alpha/2,n-1}$

If critical region satisfies reject Ho

Step#5

Calculation:

$\mu=, s =, \overline{X} =, n =,$

$$t = \frac{\overline{X}-\mu}{s/\sqrt{n}}$$

Step#6

Conclusion:

Hypothesis testing of proportion

There are 6 basic steps of hypothesis testing for population mean:

1) State the claim & opposite to the claim
2) Level of Significance
3) Test Statistics
4) Critical region
5) Calculation
6) Conclusion

Roadmap of proportion testing

Hypothesis	Test Statistics	Critical region
H_1: P>a	$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$	$Z > Z_\alpha$ If critical region satisfies reject Ho
H_1: P<a		$Z < -Z_\alpha$ If critical region satisfies reject Ho
H_1: P≠a		$Z > Z_{\alpha/2}$ or $Z < -Z_{\alpha/2}$ If critical region satisfies reject Ho

Exercise: 8.4

Question#5 **Home Ownership** A recent survey found that 68.6% of the population own their homes. In a random sample of 150 heads of households, 92 responded that they owned their homes. At the α= 0.01 level of significance, does that suggest a difference from the national proportion?

Step#1

Ho:

H_1:

Step#2

Level of significance: α

Step#3

Test Statistics

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$$

Step#4

Critical region

$Z > Z_\alpha$

$Z < -Z_\alpha$

$Z > Z_{\alpha/2}$ or $Z < -Z_{\alpha/2}$

If critical region satisfies reject Ho

Step#5

Calculation:

P=, $Q =$, $\hat{p} =$, $n =$,

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$$

Step#6

Conclusion:

Question#7 **7. Overweight Children** Health issues due to being overweight affect all age groups. Of children and adolescents 6–11 years of age, 18.8% are found to be overweight. A school district randomly sampled 130 in this age group and found that 20 were considered overweight. At $\alpha = 0.05$ is this less than the national proportion?

Step#1

Ho:

H_1:

Step#2

Level of significance: α

Step#3

Test Statistics

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$$

Step#4

Critical region

$Z > Z_\alpha$

$Z < -Z_\alpha$

$Z > Z_{\alpha/2}$ or $Z < -Z_{\alpha/2}$

If critical region satisfies reject Ho

Step#5

Calculation:

P=, $Q =, \hat{p} =, n =,$

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$$

Step#6

Conclusion:

Question#19: **Football Injuries** Areport by the NCAAstates that 57.6% of football injuries occur during practices.Ahead trainer claims that this is too high for his conference, so he randomly selects 36 injuries and finds that 17 occurred during practices. Is his claim correct, at α=0.05?

Step#1

Ho:

H_1:

Step#2

Level of significance: α

Step#3

Test Statistics

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$$

Step#4

Critical region

$Z>Z_{\alpha}$

$Z<-Z_{\alpha}$

$Z>Z_{\alpha/2}$ or $Z<-Z_{\alpha/2}$

If critical region satisfies reject Ho

Step#5

Calculation:

P=, $Q =, \hat{p} =, n =,$

$$z = \frac{\hat{p}-P}{\sqrt{\frac{PQ}{n}}}$$

Step#6

Conclusion: